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# Loop 2: Advanced Repetition

## Programming Logic · Week 10

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# Learning Outcomes

By the end of this session, you can combine and nest loops to solve repetitive problems.

**Explain**

When to choose FOR, WHILE, or DO-WHILE

**Write**

Sentinel-controlled and nested loops

**Design**

Trace tables for nested loops

```
DO
    INPUT pin
WHILE pin != 1234
```

# Choosing a Loop

## FOR

**Known count**

Number of rounds is known in advance.

## WHILE

**Condition first**

Checks first, so it may run zero times.

## DO-WHILE

**Body first**

Runs at least once, then checks.

# DO-WHILE Loop

```
DO
    INPUT score
WHILE score < 0 OR score > 100
OUTPUT score
```

DO-WHILE runs the body first, then checks the condition. It always runs at least once, which suits input validation.

```
total = 0
INPUT n
WHILE n != 0 DO
    total = total + n
    INPUT n
END WHILE
OUTPUT total
```

# Sentinel Loop

A sentinel is a special value (here 0) that tells the loop to stop. Read the first value before the loop and the next one at its end.

```
FOR row = 1 TO 3 DO
    FOR col = 1 TO 4 DO
        OUTPUT "*"
    END FOR
    NEWLINE
END FOR
```

# Nested Loop

For every round of the outer loop, the inner loop completes all of its rounds. This prints 3 rows of 4 stars.

# Tracing Nested Loops

### Iterations

| i | j |
|---|---|
| 1 | 1 |
| 1 | 2 |
| 2 | 1 |
| 2 | 2 |
| 3 | 1 |
| 3 | 2 |

### Output

| (i, j) | i * j |
|---|---|
| 1, 1 | 1 |
| 1, 2 | 2 |
| 2, 1 | 2 |
| 2, 2 | 4 |
| 3, 1 | 3 |
| 3, 2 | 6 |

### Counting Rounds

| Loop | Runs |
|---|---|
| Outer | 3 times |
| Inner | 2 per outer |
| Total | 3 x 2 = 6 |
| Order | inner first |
| Rows | outer loop |
| Columns | inner loop |

# Common Loop Pitfalls

Mistake 1

## Same counter

Reusing i for both loops makes them interfere.

Mistake 2

## Missing read

Without the second INPUT inside the loop, it never ends.

Mistake 3

## Sentinel sum

The stop value 0 gets added to the total by mistake.

Mistake 4

## Bad nesting

The inner loop must sit completely inside the outer loop.

Mistake 5

## No reset

Reset inner counters inside the outer loop, not before it.

Mistake 6

## Many rounds

Nested loops multiply: 100 x 100 is 10,000 rounds.

# Student Scores Problem

Input student scores until the user enters -1. Reject scores outside 0 to 100 and ask again.

**Tasks**

- Display how many valid scores were entered
- Display the total and the average
- Display the highest score
- Bonus: print a triangle of N rows using nested loops

**Deliverables**

- Pseudocode for the whole program
- Trace table for the inputs 80, 90, 70, -1

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